MATH 3312, SPRING 2026: SOLUTIONS TO HOMEWORK 8

(1) Suppose that R is an integral domain. Show that every (non-zero) prime element $r \in R$ is irreducible.

Solution. If $r = st$ with s and t both non-zero and non-invertible, then the obvious fact that $st \in (r)$, combined with the assumption that (r) is a prime ideal, implies that one of s and t must be in (r) (why?). Suppose that $s \in (r)$: then we get $s = ru$ for some $u \in R$. But then we have $r = st = rut$, which tells us that $r(1 - ut) = 0$. Since R is an integral domain, this can only happen if $1 - ut = 0$, which implies that t is invertible.

Definition 1. The **sum** $I_1 + I_2 \leq R$ of two ideals I_1 and I_2 is the ideal consisting of elements of the form $x + y$ with $x \in I_1$ and $y \in I_2$.

Definition 2. Two ideals $I_1, I_2 \leq R$ in a commutative ring R are **relatively prime** if $I_1 + I_2 = R$.

(2) Let R be a PID and suppose that $q_1, q_2 \in R$ are two irreducible elements generating *distinct* ideals. Show that, for any $n_1, n_2 \geq 1$, the ideals $(q_1^{n_1}), (q_2^{n_2})$ are relatively prime.

Solution. Since R is a PID, the sum $(q_1^{n_1}) + (q_2^{n_2})$ is once again principal. Suppose that it is equal to (r). Then since the sum of two ideals contains both individual ideals (why?), we have $q_1^{n_1} \in (r)$ and $q_2^{n_2} \in (r)$. This can happen only if $r \mid q_1^{n_1}$ and $r \mid q_2^{n_2}$. In other words, r is a common factor of both these prime powers. However, by unique factorization, this means that every prime factor q of r differs from both q_1 and q_2 by invertible elements (why?). That is, we have $q = u_1 q_1$ and $q = u_2 q_2$ where $u_1, u_2 \in R$ are invertible. But then we would have $(q) = (q_1) = (q_2)$ as ideals. But this is not possible, since q_1 and q_2 generate *distnct* prime ideals. This means that r has no prime factors, which can only happen if r is invertible (why?). This is the same as saying that $(r) = R$.

Definition 3. If $I_1, I_2 \leq R$ are two ideals, then their **product** $I_1 I_2$ is the ideal generated by elemens of the form xy with $x \in I_1$ and $y \in I_2$.

(3) (Chinese Remainder Theorem: General version) Suppose that $I_1, I_2 \leq R$ are two relatively prime ideals.

(a) Show that the ring homomorphism

$$R \xrightarrow{r \mapsto (r + I_1, r + I_2)} R/I_1 \times R/I_2$$

is surjective.

Solution. As the hint below says, we have $j_1 \in I_1$ and $j_2 \in I_2$ such that $j_1 + j_2 = 1$. This means that $j_2 + I_1 = 1 + I_1$ and $j_1 + I_2 = 1 + I_2$. But then, for any $r_1, r_2 \in R$, let's look at $r = r_2 j_1 + r_1 j_2$: In R/I_1, we have (since $r_2 j_1 \in I_1$)

$$r + I_1 = r_2 j_1 + r_1 j_2 + I_1 = r_1 j_2 + I_1 = (r_1 + I_1)(j_2 + I_1) = (r_1 + I_1)(1 + I_1) = r_1 + I_1.$$

Similarly, in R/I_2, we have

$$r + I_2 = r_2 j_1 + r_1 j_2 + I_2 = r_2 + I_2.$$

This tells us that $r \in R$ maps to $(r_1 + I_1, r_2 + I_2)$. Since r_1, r_2 were arbitrary, tells us that the ring homomorphism is surjective.

(b) Show that the kernel of the map in (a) is $I_1 I_2 = I_1 \cap I_2$.

Solution. The kernel consists of elements r such that $r + I_1 = I_1$ and $r + I_2 = I_2$. This can happen exactly when $r \in I_1$ and $r \in I_2$, which is of course saying that we have $r \in I_1 \cap I_2$.
The main thing to do now is to show that $I_1 I_2 = I_1 \cap I_2$. One containment is general: If $x \in I_1$ and $y \in I_2$, then xy is in both I_1 and I_2 (why?). Since $I_1 I_2$ is generated by elements of the form xy, this tells us that all the elements of $I_1 I_2$ are contained in $I_1 \cap I_2$.
For the other containment, we need the relative primeness: Suppose that $r \in I_1 \cap I_2$. Then we have $r = r \cdot 1 = r(j_1 + j_2) = rj_1 + rj_2$. Now $rj_1 = j_1 r \in I_1 I_2$, because $j_1 \in I_1$ and $r \in I_2$. Similarly, $rj_2 \in I_1 I_2$. This means that the sum $r = rj_1 + rj_2$ is also in $I_1 I_2$.

(c) Show more generally that if $I_1, \ldots, I_m \leq R$ are ideals such that, for any $r \neq s$, I_r and I_s are relatively prime, then we have an isomorphism[1]

$$R/(I_1 \cdots I_m) \xrightarrow{\simeq} R/I_1 \times \cdots \times R/I_m.$$

Solution. We do this by induction on $m \geq 2$. The base case of $m = 2$ is what we just looked at. If $m > 2$ and the result is true for $m - 1$ pairwise relatively prime ideals, then we have

$$R/(I_1 \cdots I_{m-1}) \xrightarrow{\simeq} R/I_1 \times \cdots \times R/I_{m-1}.$$

Now, I claim that $I_1 \cdots I_{m-1}$ and I_m are relatively prime. To see this, for each $s = 1, \ldots, m-1$, choose $j_s \in I_s$ and $j_{s,m} \in I_m$ such that $j_s + j_{s,m} = 1$ (why can you make such a choice?). But then we can consider the product

$$(j_1 + j_{1,m})(j_2 + j_{2,m}) \cdots (j_{m-1} + j_{m-1,m}) = 1$$

Each factor is 1 so the product is of course 1. But if we expand out the product on the left, then the first term is $j_1 j_2 \cdots j_{m-1}$, which is in $I_1 I_2 \cdots I_{m-1}$, and every other product involves an element of I_m, and so belongs to I_m. In other words, we can write 1 as a sum of an element of $I_1 I_2 \cdots I_{m-1}$ and an element of I_m. This means exactly that these two ideals are relatively prime.
But now we can apply the two ideal version of the Chinese Remainder Theorem and our inductive hypothesis, so see that

$$R/(I_1 \cdots I_m) \xrightarrow{\simeq} R/(I_1 \cdots I_{m-1}) \times R/I_m \xrightarrow{\simeq} R/I_1 \times \cdots \times R/I_{m-1} \times R/I_m.$$

(d) Suppose that R is a PID with $q_1, \ldots, q_m \in R$ irreducible elements generating *distinct* ideals. Show that, for any $n_1, \ldots, n_m \geq 1$, we have

$$R/(q_1^{n_1} \cdots q_m^{n_m}) \xrightarrow{\simeq} R/(q_1^{n_1}) \times \cdots \times R/(q_m^{n_m}).$$

Solution. Simply combine the previous part with problem (2).

Hint: For (a), use the fact that we can write $1 = j_1 + j_2$ with $j_1 \in I_1$ and $j_2 \in I_2$. For (b), the main thing is to show the equality $I_1 I_2 = I_1 \cap I_2$, and this also follows from the same observation.

Definition 4. Suppose that $A \in M_n(k)$ is conjugate to a matrix in rational canonical form corresponding to polynomials $f_1(x) \mid \cdots \mid f_m(x)$. Then the **minimal polynomial** of A, $\min_A(x)$ is the polynomial $f_m(x)$, while the **characteristic polynomial** of A, $\mathrm{ch}_A(x)$ is the polynomial $f_1(x) \cdots f_m(x)$.

(4) Show that the following are equivalent for a matrix $A \in M_n(k)$ and $\lambda \in k$:
  (a) λ is a zero for $\min_A(x)$.
  (b) λ is a zero for $\mathrm{ch}_A(x)$.
  (c) The k-linear transformation $k^n \xrightarrow{v \mapsto T_A(v) - \lambda \cdot v} k^n$ is not an isomorphism.
  (d) There exists a non-zero vector $v \in k^n$ such that $Av = \lambda v$.
  Hint: Think about the k-pair V_A as a $k[x]$-module: What does multiplication by $x - \lambda$ do in terms of the expression for V_A given by the rational canonical form?

[1]The product of multiple ideals is defined iteratively.

Solution. First, note that (a) and (b) are equivalent, since any zero for $\mathrm{ch}_A(x)$ will be a zero for one of the $f_i(x)$, and hence for $\min_A(x)$, which is divisible by all of the invariant factors.

Statements (c) and (d) are also equivalent: They are both saying that $v \mapsto Av - \lambda v$ is not injective.

The thing to show is that (a) and (c) are equivalent. The main point here is the following observation: We have an isomorphism of k-pairs

$$V_A \simeq V_{f_1} \times \cdots \times V_{f_m}$$

This means that we can translate questions about the action of $T_A - \lambda I_n$ on the left hand side into questions about multiplication by $x - \lambda$ on the right hand side: Indeed, the isomorphism of k-pairs matches up $T_A - \lambda I_n$ with precisely the linear transformation given by multiplpciation by $x - \lambda$.

This means that the equivalence of (a) and (c) is now reduced to the following statement: λ is a zero for $f(x) \in k[x]$ if and only if multiplication by $x - \lambda$ on $k[x]/(f(x))$ is *not injective*.

This is a concrete assertion. Every element of $k[x]/(f(x))$ looks like $r(x) + (f(x))$ where $\deg r(x) < \deg f(x)$. Saying that $x - \lambda$ kills such an element is equivalent to saying that

$$(x - \lambda)r(x) = f(x)g(x) \in (f(x))$$

for some $g(x) \in k[x]$. By degree considerations, this is basically equivalent to saying that we can find $r(x)$ of degree $\deg f(x) - 1$ such that $(x - \lambda)r(x) = f(x)$. This in turn is equivalent to saying that $f(\lambda) = 0$, as desired.

Definition 5. A $\lambda \in k$ satisfying the equivalent conditions of the previous problem is called an **eigenvalue** for A.

If $p(x) = a_0 + \cdots + a_{d-1}x^{d-1} + a_d x^d \in k[x]$ is a polynomial and $A \in M_n(k)$, we set

$$p(A) = a_0 I_n + \cdots + a_{d-1}A^{d-1} + a_d A^d \in M_n(k).$$

(5) Show that the following are equivalent:
 (a) $p(A) = 0 \in M_n(k)$.
 (b) $\min_A(x) \mid p(x)$.

Conclude that we always have $\mathrm{ch}_A(A) = 0$: this is the **Cayley-Hamilton theorem**.

This explains the term minimal polynomial*: It is the smallest polynomial that kills A.*

Hint: You want to use the fact that

$$V_A \simeq k[x]/(f_1(x)) \times \cdots \times k[x]/(f_m(x))$$

and note that multiplication by A on the left corresponds to multiplication by x on the right.

Solution. The hint is of course the same observation that we used in the previous problem. As it says, we are basically confronted with the following question: When is multiplication by $p(x)$ the zero linear transformation on $k[x]/(f(x))$? A moment's thought shows that this is the case precisely when $p(x) \in (f(x))$! Therefore, multiplication by $p(x)$ on

$$k[x]/(f_1(x)) \times \cdots \times k[x]/(f_m(x))$$

is zero precisely when we have $p(x) \in (f_m(x)) = (\min_A(x))$. This translates to saying that $p(A)$ is the zero matrix exactly when $p(x)$ is divisible by $\min_A(x)$.

(6) Let $M \leq \mathbb{Z}[i]^3$ be the $\mathbb{Z}[i]$-submodule generated by

$$(1 + i, 1 - i, 2), (2 + i, 2 - i, 5), (1 + 2i, 1 - 2i, 5).$$

Find the invariant factor and elementary divisor decomposition for $\mathbb{Z}[i]^3/M$.

Hint: You might want to show that $1 \pm i$, $2 \pm i$, $1 \pm 2i$ are all prime elements, though they only generate three distinct ideals among them.

Solution. We did a lot of the work in Lecture 26. We ended with the matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2i & -4i \\ 0 & 1+3i & 2-6i \end{bmatrix}$$

It remains to reduce the bottom 2×2-matrix into SNF. First, note that the gcd of the entries of this matrix is the same as the gcd between $2i$ and $1+3i$ (why?). To compute this, we do long division as usual:

$$\begin{aligned} 1+3i &= 1 \cdot 2i + (1+i) \\ 2i &= (1+i)(1+i) + 0 \end{aligned}$$

This tells us that $1+i$ is the gcd of $2i$ and $1+3i$, and it also tells us that we have

$$1+i = (-1) \cdot 2i + 1 \cdot (1+3i) \Rightarrow 1 = (-1) \cdot \frac{2i}{1+i} + 1 \cdot \frac{1+3i}{1+i} = (-1) \cdot (1+i) + 1 \cdot (2+i).$$

We now can multiply the 2×2 block by the invertible matrix $\begin{bmatrix} -1 & 1 \\ -(2+i) & 1+i \end{bmatrix}$ to get

$$\begin{bmatrix} -1 & 1 \\ -(2+i) & 1+i \end{bmatrix} \cdot \begin{bmatrix} 2i & -4i \\ 1+3i & 2-6i \end{bmatrix} = \begin{bmatrix} 1+i & 2-2i \\ 0 & 4+4i \end{bmatrix}$$

At this point, we can use a column operation to get rid of the $2-2i$ in the top right, since we already know that $1+i$ must divide it (why?). All of this now leaves us with the matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1+i & 0 \\ 0 & 0 & 4(1+i) \end{bmatrix}$$

So we can conclude that $\mathbb{Z}[i]^3/M$ has invariant factors $1+i, 4(1+i)$. To find the elementary divisors, now actually note that—as ideals—we have $(2) = ((1+i)^2)$, which shows that $(4(1+i)) = ((1+i)^4(1+i)) = ((1+i)^5)$. This means that the elementary divisors are actually the *same* as the invariant factors in this case: Everything involved is a power of $(1+i)$ as an ideal.

(7) A polynomial $f(x) \in \mathbb{Z}[x]$ is **primitive** if its coefficients have gcd 1. For example, $2x+3$ is primitive, but $2x+4$ is not.

(a) Show that $f(x) \in \mathbb{Z}[x]$ is primitive if and only if, for every prime p, the image of $f(x)$ in $\mathbb{F}_p[x]$ is non-zero.

Solution. Saying that the coefficients of $f(x)$ have no common factors is the same as saying that there is no prime p that divides all the coefficients, which is the same as saying that, when we look at the coefficients mod p, at least one coefficient is always non-zero.

(b) Use this to show that the product of two primitive polynomials is once again primitive. *Hint: $\mathbb{F}_p[x]$ is an integral domain.*

Solution. If $f(x)$ and $g(x)$ are both non-zero mod every p, then their product is also non-zero mod every p, as per the hint.

(c) Suppose that $f(x) \in \mathbb{Q}[x]$ is a non-zero polynomial. Show that there exists a positive rational number $r \in \mathbb{Q}$ such that $r \cdot f(x)$ is a primitive polynomial in $\mathbb{Z}[x]$.

Solution. First, choose an integer m such that $m \cdot f(x) \in \mathbb{Z}[x]$: you can do this by clearing all the denominators of the coefficients. Now, we can look at the gcd n of all the coefficients of $m \cdot f(x)$. The multiple $\frac{m}{n} \cdot f(x)$ will then be the primitive polynomial we seek (why?).

(d) Suppose that $f(x) \in \mathbb{Z}[x]$ is a polynomial with leading coefficient 1, and that $h(x), g(x) \in \mathbb{Q}[x]$ are also polynomials with leading coefficient 1 such that $f(x) = h(x)g(x)$. Show that $h(x), g(x) \in \mathbb{Z}[x]$. *Hint: Apply the previous part to $h(x)$ and $g(x)$ and see what that tells you.*

Solution. By the previous part we can find rational numbers r_1, r_2 such that $r_1 h(x)$ and $r_2 g(x)$ are both primitive in $\mathbb{Z}[x]$. Since $h(x)$ and $g(x)$ have leading coefficient 1, this is only possible if r_1 and r_2 are both integers (why?) By part (b), this implies that $(r_1 h(x))(r_2 g(x))$ is primitive. But this is equal to $r_1 r_2 f(x)$.

Now, since $f(x)$ has leading coefficient 1, the only way for $r_1 r_2 f(x)$ to be an integer polynomial is to have $r_1 r_2 = 1$. Since r_1 and r_2 are both integers, this is only possible if either $r_1 = r_2 = 1$ or $r_1 = r_2 = -1$. In both cases, it implies that $h(x)$ and $g(x)$ are already integer polynomials.

The fourth statement is usually called Gauss's lemma.